

Congruent modular forms and anticyclotomic Iwasawa theory

Dac-Nhan-Tam Nguyen^{1, 2}

¹Department of Mathematics, University of British Columbia, 1984 Mathematics Road, Vancouver, V6T 1Z2, British Columbia, Canada.

²ORCID: 0000-0002-3117-5028.

Contributing authors: tamnguyen@math.ubc.ca;

Abstract

Let p be an odd prime. Consider normalized newforms f_1, f_2 that both satisfy the Heegner hypothesis for an imaginary quadratic field K and suppose that they induce isomorphic residual Galois representations. In the work of Greenberg-Vatsal [1] and Emerton-Pollack-Weston [2], the authors compare the cyclotomic Iwasawa μ and λ -invariants of f_1 and f_2 . We extend this to the anticyclotomic indefinite setting by comparing the BDP p -adic L -functions attached to f_1 and f_2 . Using this comparison, we obtain arithmetic implications for both generalized Heegner cycles and the Iwasawa main conjecture.

Keywords: Anticyclotomic Iwasawa theory, congruent modular forms, p -adic L -functions, Heegner cycles.

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1 Introduction

Let $f \in S_{2r}(\Gamma_0(N))^{\text{new}}$ be a normalized newform of weight $2r$ and level N that is an eigenform for all Hecke operators. Fix an odd prime $p \nmid N$. We may attach to f a p -adic Galois representation V_f and let

$$\rho_f : G_{\mathbb{Q}} \rightarrow \text{Aut}(V_f(r))$$

be its self-dual Artin twist. We will denote by $\bar{\rho}_f$ the associated semisimplified residual representation.

Let $K/\mathbb{Q}$ be an imaginary quadratic field of discriminant $-D_K$ and let $p > 2$ be a rational prime that split in K as $(p) = \mathfrak{p}\bar{\mathfrak{p}}$. We define the following hypothesis for $f \in S_{2r}(\Gamma_0(N))^{\text{new}}$

$$\begin{cases} p \nmid 2(2r-1)!N\phi(N), \\ \text{every prime } \ell \mid N \text{ is split in } K/\mathbb{Q}. \end{cases} \quad (\text{Heeg})$$

The second condition is known as the strong Heegner hypothesis. In such a setting, we may define the BDP anticyclotomic p -adic L -function $\mathcal{L}_{\mathfrak{p}}(f)$ attached to f in the sense of [3–5]. This paper closely follows [5], whose construction of the p -adic L -function originates from [4]. This p -adic L -function is defined as an element of the Iwasawa algebra $\mathcal{W}[[\Gamma_K^-]]$ where $\mathcal{W}$ is a certain finite extension of the completed maximal unramified extension $\widehat{\mathbb{Q}_p^{nr}}$ and Γ_K^- is the Galois group of the anticyclotomic extension over K .

It is natural to ask how the Iwasawa μ and λ -invariants of $\mathcal{L}_{\mathfrak{p}}(f_1)$ and $\mathcal{L}_{\mathfrak{p}}(f_2)$ differ for newforms f_1 and f_2 whose residual representations are isomorphic. This type of question was first studied in [1] over the cyclotomic extension, which was then generalized in [2]. The papers [6], [7], [8] give analogous results in the definite anticyclotomic setting.

In the indefinite anticyclotomic setting, congruences between the BDP p -adic L -functions have been studied in [9] for the weight 2 case. In this setting, Kriz-Li studied the logarithms of Heegner points twisted by unramified characters which are interpolated by the BDP p -adic L -functions (see [10, Theorem 3.9]). The results in this paper can be seen as generalizations of [9, 10] to forms of higher weights and generalized Heegner cycles. The techniques in this paper differ from [9] and as a result require fewer hypotheses. Moreover, for modular forms that are residually isomorphic with respect to an arbitrary prime power, we are able to show congruences between their p -adic L -functions with respect to the same prime power (see Theorem 5.10). In a paper by Castella *et al.* [11], the authors use congruence methods to acquire new instances of the anticyclotomic Iwasawa main conjecture at Eisenstein primes. Their work can be seen as an extension of [1, Theorem (1.3)] to the BDP p -adic L -function whereas our work (in particular, Theorem 7.5) extends [1, Theorem (1.4)].

In [10], the authors study congruences by looking at the stabilizations of f_1 and f_2 at various primes ℓ . These stabilizations are based on Hecke operators that act on classical modular form $f \in S_{2r}(\Gamma_0(N))$ via $f(q) \mapsto f(q^\ell)$. To study how the anticyclotomic p -adic L -function varies, this paper introduces some suitable moduli interpretations of these Hecke operators in the context of Igusa schemes in Section 5.2, which will be relevant for the construction via Serre-Tate coordinates as defined in [4, 5]. We also note that the moduli interpretations of some Hecke operators attached to the prime p are discussed in [12, Section 4.1.10].

We also explore arithmetic implications for Heegner cycles in Section 6, as well as the anticyclotomic Iwasawa main conjecture in Section 7. We now state the main results of this paper.

Suppose that $f_1 \in S_{2r_1}(\Gamma_0(N_1))^{\text{new}}, f_2 \in S_{2r_2}(\Gamma_0(N_2))^{\text{new}}$ are normalized Hecke eigenforms whose coefficients lie in some number field L . We denote by L_p the completion of L with respect to a fixed prime above p . Suppose that the induced semi-simplified mod ϖ^m Galois representations $\bar{\rho}_{f_1}, \bar{\rho}_{f_2} : \text{Gal}(\bar{L}/L) \rightarrow GL_2(\mathcal{O}_{L_p}/\varpi^m \mathcal{O}_{L_p})$ are isomorphic, where ϖ is the uniformizer of $\mathcal{O}_{L_p}$. Let $\mathcal{W}$ be the ring of integers of a finite extension of $\widehat{\mathbb{Q}_p^{nr}}$ containing L .

Theorem A (Theorem 5.10). *Suppose that both f_1, f_2 satisfy hypothesis (Heeg) for $K/\mathbb{Q}$. We may write $(N_1) = \mathfrak{N}_1 \bar{\mathfrak{N}}_1, (N_2) = \mathfrak{N}_2 \bar{\mathfrak{N}}_2$ as ideals in $\mathcal{O}_K$. For each prime $\ell \mid N_1 N_2$, let $v \mid \mathfrak{N}_1 \bar{\mathfrak{N}}_2$ be the corresponding prime above ℓ . Then the following congruence holds:*

$$\prod_{\ell \mid N_1 N_2} \mathcal{P}_{\bar{v}}(f_1) \mathcal{L}_{\mathfrak{p}}(f_1) \equiv \prod_{\ell \mid N_1 N_2} \mathcal{P}_{\bar{v}}(f_2) \mathcal{L}_{\mathfrak{p}}(f_2) \pmod{\varpi^m \mathcal{W}[\Gamma_K^-]},$$

where for each $\bar{v}$, $\mathcal{P}_{\bar{v}}(f_1)$ and $\mathcal{P}_{\bar{v}}(f_2)$ are defined in Definition 5.7. Moreover, one has the following:

1. $\mu(\mathcal{L}_{\mathfrak{p}}(f_1)) = 0$ if and only if $\mu(\mathcal{L}_{\mathfrak{p}}(f_2)) = 0$.
2. Assuming that $\mu(\mathcal{L}_{\mathfrak{p}}(f_1)) = \mu(\mathcal{L}_{\mathfrak{p}}(f_2)) = 0$,

$$\sum_{\ell \mid N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_1)) + \lambda(\mathcal{L}_{\mathfrak{p}}(f_1)) = \sum_{\ell \mid N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_2)) + \lambda(\mathcal{L}_{\mathfrak{p}}(f_2)).$$

Notation. Throughout this paper, we fix embeddings $\iota_\infty : \bar{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and $\iota_p : \bar{\mathbb{Q}} \hookrightarrow \mathbb{C}_p$. Let $v_p(\cdot)$ be the normalized additive valuation on $\mathbb{C}_p$ for which $v_p(p) = 1$.

For each number field F , the embedding ι_p determines a choice of inclusion $F \subset \mathbb{C}_p$, or equivalently a prime in F above p . We assume that this choice gives rise to the prime $\mathfrak{p}$ in K that is consistent with the splitting $p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}$ given in the Introduction. We will denote by F_p the completion of F with respect to the prime induced by ι_p . We will also denote by $\mathbb{A}_F$ the adeles of F and $\hat{F}$ the finite adeles. Moreover, let $F_\infty := \prod_{v \mid \infty} F_v$.

Let $K[c]$ be the ring class field of conductor c over K , and write $K[p^\infty]$ for $\bigcup_{n \geq 0} K[p^n]$. Denote by $\bar{\Gamma}$ the Galois group of $K[p^\infty]/K$, and let Γ_K^- be the maximal pro- p quotient so that $\Gamma_K^- = \text{Gal}(K_\infty/K)$ is the Galois group of the anticyclotomic extension $K_\infty = \bigcup_{n \geq 0} K_n$ over K . Let $\text{rec}_{\mathfrak{p}} : \mathbb{Q}_p^\times = K_{\mathfrak{p}}^\times \rightarrow \text{Gal}(K^{ab}/K) \rightarrow$

$\text{Gal}(K[p^\infty]/K)$ be the local reciprocity map. We also write $K(\mathfrak{p}^\infty)$ for the ray class field of conductor $\mathfrak{p}^\infty$, and $K[c](\mathfrak{p}^\infty)$ for the compositum of $K[c]$ and $K(\mathfrak{p}^\infty)$.

2 Geometric and p -adic modular forms

We follow the expositions in Brakočević [4] and Castella-Hsieh [5] and recall the definitions of (geometric, p -adic) modular forms of levels $\Gamma_0(N)$ and $\Gamma_1(N)$. The main references for this section are [13], [12, Section 3].

Let S denote a p -adic ring (which is also a $\mathbb{Z}_p$ -algebra) and R denote some p -adic S -algebra. For an integer N , we let μ_N be the group scheme of the N -th roots of unity and let $A[N]$ be the group scheme of the N -torsion points of an abelian variety A .

Consider the isomorphism classes of triples $[(A, \eta_N, \omega)_{/R}]_{\simeq}$, where A/R is an elliptic curve and $\eta_N : \mu_N \rightarrow A[N]$ is the $\Gamma_1(N)$ -level structure and $\omega \in H^0(A/R, \underline{\Omega}_{A/R}^1)$ is a differential 1-form. The functor classifying such triples is representable by an affine scheme $\mathcal{M}_{\Gamma_1(N)}$ defined over $\mathbb{Z}[1/6N]$ [12, Theorem 3.1].

Definition 2.1. ([12, Section 3.2.3]) For each S -algebra R , consider the set of all triples $[(A, \eta_N, \omega)_{/R}] \in \mathcal{M}_{\Gamma_1(N)}(R)$. A geometric modular form f of weight k and level $\Gamma_1(N)$ over R is a rule assigning to such every triple $(A, \eta, \omega)_{/R}$ a value $f(A, \eta, \omega) \in R$ satisfying the following:

1. $f(A, \eta, \omega) = f(A', \eta', \omega')$ if $(A, \eta, \omega) \simeq (A', \eta', \omega')$ over R .
2. For any S -algebra homomorphism $\phi : R \rightarrow R'$, we have

$$f((A, \eta, \omega) \otimes_R R') \simeq \phi(f(A, \eta, \omega))$$

3. $f(A, \eta, \lambda\omega) = \lambda^{-k} f(A, \eta, \omega)$ for any $\lambda \in R^\times$.
4. Let $\text{Tate}(q)$ be the Tate curve $\mathbf{G}_m/q^{\mathbb{Z}}$ over $\mathbb{Z}((q))$, equipped with a level structure η and a choice of differential ω . Then $(\text{Tate}(q), \eta, \omega)$ is defined over $S[\mu_d]((q^{1/d}))$ for some $d \mid N$, and we impose that $f(\text{Tate}(q), \eta, \omega) \in S[\mu_d][[q^{1/d}]]$ for every such $(\text{Tate}(q), \eta, \omega)$.

Moreover, we say that f is of level $\Gamma_0(N)$ if it also satisfies

5. $f((A, \eta_N \circ b, \omega)_{/R}) = f(A, \eta_N, \omega)$ for any $b \in (\mathbb{Z}/N\mathbb{Z})^\times$ with the canonical action of $(\mathbb{Z}/N\mathbb{Z})$ on μ_N [4, Section 3.1].

We define the q -expansion of f as $f(\text{Tate}(q), \eta_{\text{can}}, du/u) \in S[[q]]$, where $\eta_{\text{can}} : \mu_N \oplus \mu_{p^\infty} \hookrightarrow \mathbf{G}_m \rightarrow \mathbf{G}_m/q^{\mathbb{Z}}$ is the canonical level structure, and u is the canonical parameter of $\mathbf{G}_m = \text{Spec}(\mathbb{Z}[u, u^{-1}])$.

To define p -adic modular forms, we first introduce the Igusa scheme $\text{Ig}(N)/\mathbb{Z}_p$, which is the moduli space parametrizing isomorphism classes of elliptic curves with $\Gamma_1(Np^\infty)$ -structure. More precisely, for each $\mathbb{Z}_p$ -algebra R , $\text{Ig}(N)(R)$ is the isomorphism classes of tuples $[(A, \eta)_{/R}]_{\simeq}$ where A/R is an elliptic curve and $\eta = \eta_N \oplus \eta_p : \mu_N \oplus \mu_{p^\infty} \hookrightarrow A[N] \oplus A[p^\infty]$ is an immersion of group schemes ([12, Section 3.2.7]).

Definition 2.2. ([12, Section 3.2.9], [5, Section 2.1]) Denote by $S_m := S/p^m S$. We define the space of p -adic modular forms of level $\Gamma_1(N)$ over S , denoted $V_p(\Gamma_1(N), S)$, as

$$V_p(\Gamma_1(N), S) = H^0(\widehat{\text{Ig}}(N), \mathcal{O}_{\widehat{\text{Ig}}(N)/S}) = \varprojlim_m H^0(\text{Ig}(N), \mathcal{O}_{\text{Ig}(N)/S_m}),$$

where $\widehat{\text{Ig}}(N)$ is the formal completion of $\text{Ig}(N)$. In particular, f is a function assigning to each $[(A, \eta)_{/R}] \in \text{Ig}(N)(R)$ a value $f(A, \eta) \in R$, and they satisfy the following conditions:

1. $f((A, \eta)_{/R}) = f((A', \eta')_{/R})$ if $(A, \eta)_{/R} \simeq (A', \eta')_{/R}$.
2. For any continuous homomorphisms of S -algebra $\phi : R \rightarrow R'$, we have

$$f((A, \eta) \otimes_R R') \simeq \phi(f(A, \eta))$$

3. For any level structure η_N of type $\Gamma_1(N)$ on the Tate curve $\text{Tate}(q)$, $f(\text{Tate}(q), \eta_N \oplus \eta_p^{\text{can}}) \in S[[q^{1/N}]]$, where η_p^{can} is determined by the canonical image of ζ_p via $\mathbf{G}_m \rightarrow \text{Tate}(q)$.

A p -adic modular form is said to be of weight k if $f(A, z^{-1}\eta_p, \eta_N) = z^k f(A, \eta_p, \eta_N)$ for all $z \in \mathbb{Z}_p^\times$.

A geometric modular form gives rise to a p -adic modular form in the sense of [14, (1.10.15)]: Let R be a complete local S -algebra, and let $[(A, \eta)_{/R}] \in \text{Ig}(N)(R)$. The $\Gamma_1(Np^\infty)$ -level structure $\eta = \eta_N \oplus \eta_p$ determines a map $\widehat{\eta}_p : \widehat{\mathbf{G}}_m \xrightarrow{\sim} \widehat{A}$ [14, (1.10.1)] (see also [15, Proposition 1]). This in turn defines a differential $\omega(\widehat{\eta}_p) : \text{Lie}(A) \simeq \text{Lie}(\widehat{A}) \rightarrow \text{Lie}(\widehat{\mathbf{G}}_m) = R$. One can then define the p -adic avatar $\widehat{f}$ of f ([5]) by letting $\widehat{f}(A, \eta) = f(A, \eta, \omega(\widehat{\eta}_p))$.

3 CM points

This section follows [5, Section 2]. Let K be an imaginary quadratic field of discriminant $-D_K < 0$, and suppose that p is split as $p = \mathfrak{p}\overline{\mathfrak{p}}$ in $\mathcal{O}_K$. Let $f \in S_{2r}(\Gamma_0(N))^{\text{new}}$ be a newform satisfying hypothesis (Heeg). We may write $N = \mathfrak{N}\overline{\mathfrak{N}}$ for some ideal $\mathfrak{N}$ in $\mathcal{O}_K$. For a positive integer c , let $\mathcal{O}_c := \mathbb{Z} + c\mathcal{O}_K$ be the order of conductor c in K , so that $\text{Gal}(K[c]/K) \simeq \mathcal{C}\ell(\mathcal{O}_c)$.

For each prime-to- $\mathfrak{N}\mathfrak{p}$ integral ideal $\mathfrak{a}$ of $\mathcal{O}_c$, we may attach a CM point $x_{\mathfrak{a}} = (A_{\mathfrak{a}}, \eta_{\mathfrak{a}})$. Such a point is defined over a discrete valuation ring inside $\mathcal{V} = \iota_p^{-1}(\mathcal{O}_{\mathbb{C}_p}) \cap K^{ab}$. If $\mathfrak{a} = \mathcal{O}_c$, we write (A_c, η_c) for $(A_{\mathcal{O}_c}, \eta_{\mathcal{O}_c})$. In this case, we see immediately that $A_{\mathfrak{a}} = A_c/A_c[\mathfrak{a}]$ and the isogeny $\lambda_{\mathfrak{a}} : A_c \rightarrow A_{\mathfrak{a}}$ induced by the quotient map $\mathbb{C}/\mathcal{O}_c \rightarrow \mathbb{C}/\mathfrak{a}^{-1}$ yields $\eta_{\mathfrak{a}} = \lambda_{\mathfrak{a}} \circ \eta_c$. An equivalent construction is also available in [4, Section 5.1].

If we let $\mathbb{H}$ be the complex upperhalf plane, then is a complex uniformization

$$Y_1(Np^n)(\mathbb{C}) = \text{GL}_2(\mathbb{Q})^+ \backslash \mathbb{H} \times \text{GL}_2(\widehat{\mathbb{Q}})/U_1(Np^n)$$

of complex points on the modular curve. Since the generic fiber $\text{Ig}(N)_{/\mathbb{Q}}$ is given by

$$\text{Ig}(N)_{/\mathbb{Q}} = \varprojlim_n Y_1(Np^n),$$

there is also a uniformization

$$\mathbb{H} \times \text{GL}_2(\widehat{\mathbb{Q}}) \rightarrow \text{Ig}(N)(\mathbb{C})$$

$$x = (\tau_x, g_x) \mapsto (A_x, \eta_x)$$

where (A_x, η_x) is the corresponding moduli description. We refer readers to [5, Section 2.1] for the explicit form of this map. Moreover, we will also denote the right action of $GL_2(\widehat{\mathbb{Q}})$ on $x = [(\tau_x, g_x)] \in \text{Ig}(N)(\mathbb{C})$ as

$$(\tau_x, g_x) * h := (\tau_x, g_x h).$$

Now, fix a choice of basis element ϑ for $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\vartheta$. Consider the embedding $K \hookrightarrow GL_2(\mathbb{Q})$ by the regular representation [4, 14]:

$$\rho(\alpha) \begin{pmatrix} \vartheta \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha\vartheta \\ \alpha \end{pmatrix}.$$

For the choice of ϑ given in [5, Section 2.3]:

$$\vartheta = \frac{D' + \sqrt{-D_K}}{2}, \text{ where } D' = \begin{cases} D_K & \text{if } 2 \nmid D_K \\ D_K/2 & \text{if } 2 \mid D_K \end{cases},$$

the embedding $\rho : K \hookrightarrow GL_2(\mathbb{Q})$ is of the form

$$a + b\vartheta \mapsto \begin{pmatrix} a(\vartheta + \bar{\vartheta}) + b & -a\vartheta\bar{\vartheta} \\ a & b \end{pmatrix}.$$

Tensoring with $\mathbb{A}_{\mathbb{Q}}^{(\infty)}$ gives an embedding $\rho : K^\times \backslash \widehat{K}^\times \hookrightarrow GL_2(\mathbb{Q}) \backslash GL_2(\widehat{\mathbb{Q}})$. Denote by $[\eta, g]$ the image of (η, g) under the projection $\mathbb{H} \times GL_2(\widehat{\mathbb{Q}}) \rightarrow \text{Ig}(N)(\mathbb{C})$. Then $[\eta, g] \in \text{Ig}(N)(K^{ab})$, and Shimura's reciprocity law states that

$$\text{rec}_K(a)[(\vartheta, g)] = [\vartheta, \rho(\bar{a})g]$$

where $\text{rec}_K : K^\times \backslash \widehat{K}^\times \rightarrow \text{Gal}(K^{ab}/K)$ is the geometrically normalized reciprocity law. We apply this to CM points as follows. Let $[(\vartheta, \xi_c)] \in \text{Ig}(N)(\mathbb{C})$ be the complex uniformization of the CM point $x_c := [(A_c, \eta_c)]$ for some $\xi_c \in GL_2(\widehat{\mathbb{Q}})$. For an $\mathcal{O}_c$ -ideal $\mathfrak{a}$ that is prime to $\mathfrak{N}\mathfrak{p}$, let $x_{\mathfrak{a}} = (A_{\mathfrak{a}}, \eta_{\mathfrak{a}})$ and $a \in \widehat{K}^{(cp)\times}$ be an idele such that $\mathfrak{a} = a\widehat{\mathcal{O}}_c \cap K$. Both $x_{\mathfrak{a}}$ and x_c are defined over $K[c](\mathfrak{p}^\infty)$ and $x_{\mathfrak{a}} = [(A_{\mathfrak{a}}, \eta_{\mathfrak{a}})] = [\vartheta, \rho(\bar{a})^{-1}\zeta_c] = x_c^{\sigma_{\mathfrak{a}}} \in \text{Ig}(N)(K[c](\mathfrak{p}^\infty))$ where $\sigma_{\mathfrak{a}} = \text{rec}_K(a^{-1})|_{K[c](\mathfrak{p}^\infty)} \in \text{Gal}(K[c](\mathfrak{p}^\infty)/K)$, following Shimura's reciprocity law.

4 Anticyclotomic p -adic L -functions

Let $f \in S_{2r}(\Gamma_0(N))^{\text{new}}$ be a classical normalized eigenform, which we implicitly assume to be an eigenform with respect to all Hecke operators unless otherwise stated. We will also denote by f the associated geometric modular form, and let $f^{\flat}$ be the p -depleted geometric modular form with q -expansion $f^{\flat}(q) = \sum_{p \nmid n} a_n(f)q^n$ ([5, 10, 11]).

4.1 t -expansion of p -adic modular forms

. Denote by $\widehat{\mathbb{Q}_p^{nr}}$ the completion of the maximal unramified extension of $\mathbb{Q}_p$ and let $\mathcal{W}$ be its ring of integers. Let $\mathrm{Ig}(N)_{/\mathcal{W}}$ be the Igusa scheme over $\mathcal{W}$, and let $\mathbf{x} = [(A_0, \eta)] \in \mathrm{Ig}(N)(\bar{\mathbb{F}}_p)$ where A_0 is an elliptic curve over $\bar{\mathbb{F}}_p$ and $\eta : \mu_N \oplus \mu_{p^\infty} \hookrightarrow A_0[N] \oplus A_0[p^\infty]$ is a $\Gamma_1(Np^\infty)$ -level structure. Let $\hat{S}_{\mathbf{x}}$ be the local deformation space of $\mathbf{x}$, which represents the functor

$$R \longmapsto \{\text{deformations of } A_0 \text{ to } R\}$$

for Artin local rings R with residue field $\bar{\mathbb{F}}_p$. Note that $\mathcal{W}$ is the ring of Witt vectors of $\bar{\mathbb{F}}_p$ and $\hat{S}_{\mathbf{x}}$ is a $\mathcal{W}$ -scheme [16, Section 3].

One has a natural embedding $\hat{S}_{\mathbf{x}} \hookrightarrow \mathrm{Ig}(N)_{/\mathcal{W}}$. By [16, Theorem 2.1], there is an equivalence of functors

$$\hat{S}_{\mathbf{x}} \simeq \mathrm{Hom}_{\mathbb{Z}_p}(T_p(A_0)(\bar{\mathbb{F}}_p) \otimes T_p(A_0^t)(\bar{\mathbb{F}}_p)), \hat{\mathbf{G}}_m),$$

where A_0^t is the dual of A_0 and $T_p(A_0)$, $T_p(A_0^t)$ are, respectively, the Tate modules of A_0 and A_0^t .

We denote by $q_{\mathcal{A}}$ the pairing corresponding to the isomorphism class $[\mathcal{A}/_R]$. As remarked in [5, Section 3.1], η_p determines a point $P_{\mathbf{x}} \in T_p(A_0^t)$ via the Weil pairing, which gives the canonical Serre-Tate coordinates $t : \hat{S}_{\mathbf{x}} \rightarrow \hat{\mathbf{G}}_m$ as

$$t([\mathcal{A}/_R]) = q_{\mathcal{A}}(\lambda_{\mathrm{can}}^{-1}(P_{\mathbf{x}}), P_{\mathbf{x}}),$$

together with an identification $\mathcal{O}_{\hat{S}_{\mathbf{x}}} \simeq \mathcal{W}[[t-1]]$. For a p -adic modular form $f \in V(N, \mathcal{W})$, we will denote $f(t) := f|_{\hat{S}_{\mathbf{x}}} \in \mathcal{W}[[t-1]]$.

Following [17, Sections 3.3, 3.5], we denote by $\mathrm{Meas}(\mathbb{Z}_p; \mathcal{W})$ the space of p -adic measures with values in $\mathcal{W}$. Recall the isomorphism

$$\mathrm{Meas}(\mathbb{Z}_p; \mathcal{W}) \xrightarrow{\simeq} \mathcal{W}[[t-1]]$$

given by

$$\varphi \mapsto \Phi_{\varphi}(t) = \sum_{n=0}^{\infty} \left(\int_{\mathbb{Z}_p} \binom{x}{n} d\varphi(x) \right) (t-1)^n = \int_{\mathbb{Z}_p} t^x d\varphi(x),$$

and let $df \in \mathrm{Meas}(\mathbb{Z}_p; \mathcal{W})$ be the measure corresponding to f under this isomorphism.

Following the notation of [5, p. 8], for a continuous function $\phi : \mathbb{Z}_p \rightarrow \mathcal{O}_{\mathbb{C}_p}$, we define $(f \otimes \phi)(t) \in \mathcal{O}_{\mathbb{C}_p}[[t-1]]$ by

$$(f \otimes \phi)(t) = \int_{\mathbb{Z}_p} \phi(x) t^x df = \sum_{n \geq 0} \int_{\mathbb{Z}_p} \phi(x) \binom{x}{n} df(x) \cdot (t-1)^n.$$

For a classical newform f of weight $2r$ in $S_{2r}^{\mathrm{new}}(\Gamma_0(N))$, its Fourier coefficients $\{a_n(f)\}_{n \geq 0}$ generate a number field L . We may enlarge $\mathcal{W}$ to be the ring of integers of the compositum $\widehat{\mathbb{Q}_p^{nr}} \cdot L_p$, so that both $\hat{f}$ and $\hat{f}^b$ are p -adic modular forms over $\mathcal{W}$.

Note that $\mathcal{W}$ is still a complete discrete valuation ring with residue field $\bar{\mathbb{F}}_p$. We may then define the t -expansions $\hat{f}(t), \hat{f}^b(t) \in \mathcal{W}[[t-1]]$ as above.

4.2 Hecke characters

A Hecke character $\chi : \mathbb{A}_K^\times / K^\times \rightarrow \mathbb{C}^\times$ is said to be of infinity type (m, n) if $\chi(z_\infty) = z_\infty^m \bar{z}_\infty^n$. If χ has conductor $\mathfrak{c}$, we will identify χ as a character on the ideal class group of conductor $\mathfrak{c}$ via $\psi(\mathfrak{a}) = \psi(a)$ where $a \in \mathbb{A}_K$ such that $a\hat{\mathcal{O}}_K \cap K = \mathfrak{a}$, and $a_{\mathfrak{q}} = 1$ for $\mathfrak{q} \mid \mathfrak{c}$. We write $\chi_{\mathfrak{q}}$ for the $\mathfrak{q}$ -component of χ .

Moreover, we call χ an anticyclotomic Hecke character if χ is trivial on $\mathbb{A}_{\mathbb{Q}}^\times$. For such a Hecke character χ , the p -adic avatar $\hat{\chi} : \hat{K}^\times / K^\times \rightarrow \mathbb{C}_p^\times$ is defined by $\hat{\chi}(a) = \iota_p \circ \iota_\infty^{-1}(\chi(a)) a_{\mathfrak{p}}^{-m} a_{\bar{\mathfrak{p}}}^{-n}$. We also call a p -adic character $\rho : \hat{K}^\times / K^\times \rightarrow \mathbb{C}_p^\times$ locally algebraic if $\rho = \hat{\chi}$ for some complex Hecke character χ , and define the conductor of ρ to be the conductor of χ .

For every locally algebraic character $\rho : \tilde{\Gamma} \rightarrow \mathcal{O}_{\mathbb{C}_p}^\times$, we denote by $\rho_{\mathfrak{p}}$ the character $\rho_{\mathfrak{p}} : \mathbb{Q}_p^\times \rightarrow \mathbb{C}_p^\times$ defined by $\rho_{\mathfrak{p}}(\beta) = \rho(\text{rec}_{\mathfrak{p}}(\beta))$. For a general continuous function $\rho \in \mathcal{C}(\tilde{\Gamma}, \mathcal{O}_{\mathbb{C}_p})$, we also define $\rho|[\mathfrak{a}] : \mathbb{Z}_p^\times \rightarrow \mathcal{O}_{\mathbb{C}_p}$ as $\rho|[\mathfrak{a}](x) = \rho(\text{rec}_{\mathfrak{p}}(x) \text{rec}_K(a))$. Denote by $\mathfrak{X}_{p^\infty} \subset \mathcal{C}(\tilde{\Gamma}, \mathcal{O}_{\mathbb{C}_p})$ the set of locally algebraic p -adic characters $\tilde{\Gamma} \rightarrow \mathcal{O}_{\mathbb{C}_p}^\times$.

Finally, for a continuous local character $\phi : \mathbb{Z}_q^\times \rightarrow \mathbb{C}^\times$ that necessarily factors through $(\mathbb{Z}_q/q^n\mathbb{Z}_q)^\times$ for some n , we define its Gauss sum to be $\mathfrak{g}(\phi) = \sum_{u \in (\mathbb{Z}/q^n\mathbb{Z})^\times} \phi(u) \zeta^u$, where $\zeta = e^{2\pi i/q^n}$.

4.3 Anticyclotomic p -adic L -function

For a positive integer $c = c_0 p^n$ where $\gcd(c_0, p) = 1$, let $\mathfrak{a}$ be a fractional ideal of $\mathcal{O}_c = \mathbb{Z} + c\mathcal{O}_K$ and $[(A_{\mathfrak{a}}, \eta_{\mathfrak{a}})] \in \text{Ig}(N)(K[c])$ be the corresponding CM point on the Igusa scheme discussed in Section 3. Let $\tilde{\Gamma}_K := \text{Gal}(K[p^\infty]/K)$ be the Galois group of the compositum of ring class fields of K with p^{th} -power conductor over K .

Following [5, p.12], let $\mathfrak{a} \subset \mathcal{O}_{c_0}$ be a fractional ideal prime with Np and let $t_{\mathfrak{a}}$ be the canonical Serre-Tate coordinate of $\hat{f}^b$ around the reduction $\mathbf{x}_{\mathfrak{a}} = [(A_{\mathfrak{a}}, \eta_{\mathfrak{a}})] \otimes_{\mathcal{W}} \bar{\mathbb{F}}_p$ of $[(A_{\mathfrak{a}}, \eta_{\mathfrak{a}})] \in \text{Ig}(N)(K[c_0])$. Finally, set

$$\hat{f}_{\mathfrak{a}}^b(t_{\mathfrak{a}}) := \hat{f}^b(t_{\mathfrak{a}}^{N(\mathfrak{a})^{-1} \sqrt{-D_K}^{-1}}) \in \mathcal{W}[[t_{\mathfrak{a}} - 1]],$$

where $N(\mathfrak{a}) = c^{-1} \#(\mathcal{O}_{c_0}/\mathfrak{a})$ ([5, Section 3.2]).

Definition 4.1. [5, Definition 3.7] Let $c_0 \geq 1$ be a positive integer such that $(c_0, pN) = 1$ and let ψ be an anticyclotomic Hecke character of infinity type $(r, -r)$ of conductor $c_0\mathcal{O}_K$. Define $\mathcal{L}_{\mathfrak{p}, \psi}(f)$ on $\tilde{\Gamma}$ to be the p -adic measure on $\tilde{\Gamma}$ given by

$$\mathcal{L}_{\mathfrak{p}, \psi}(f)(\rho) = \sum_{[\mathfrak{a}] \in \text{Pic}_{\mathcal{O}_{c_0}}} \psi(\mathfrak{a}) N(\mathfrak{a})^{-r} \cdot \int_{\mathbb{Z}_p^\times} \psi_{\mathfrak{p}} \rho|[\mathfrak{a}] d\hat{f}_{\mathfrak{a}}^b$$

for every continuous function $\rho : \tilde{\Gamma} \rightarrow \mathcal{O}_{\mathbb{C}_p}$. We can also view $\mathcal{L}_{\mathfrak{p}, \psi}(f)$ as an element in the semi-local ring $\mathcal{W}[[\tilde{\Gamma}]]$. It is known that $\mathcal{L}_{\mathfrak{p}, \psi}(f) \neq 0$ [5, Theorem 3.9].

For a character $\rho : \tilde{\Gamma} \rightarrow \mathcal{O}_{\mathbb{C}_p}^\times$, we define the map $\text{Tw}_\rho : \mathcal{W}[\tilde{\Gamma}] \rightarrow \mathcal{W}[\tilde{\Gamma}]$ given by $\sigma \mapsto \rho(\sigma)\sigma$ for $\sigma \in \tilde{\Gamma}$. We will denote $\mathcal{L}_\mathbf{p}(f) := \text{Tw}_{\hat{\psi}^{-1}}(\mathcal{L}_{\mathbf{p},\psi}(f))$, which takes the value

$$\mathcal{L}_\mathbf{p}(f)(\rho) = \sum_{[\mathbf{a}] \in \text{Pic}\mathcal{O}_{c_0}} N(\mathbf{a})^{-r} \cdot \int_{\mathbb{Z}_p^\times} \rho[\mathbf{a}](x) x^{-r} df_{\mathbf{a}}^b$$

for every continuous function $\rho : \tilde{\Gamma} \rightarrow \mathcal{O}_{\mathbb{C}_p}$ (see also [18, Definition 4.2]). For simplicity, we may assume that $c_0 = 1$ and $\text{Pic}\mathcal{O}_{c_0} = \text{Pic}(\mathcal{O}_K)$.

4.4 The θ operator

Let θ be the operator $t \frac{d}{dt}$ on $\mathcal{W}[[t-1]]$ and for $k < 0$ define

$$\theta^k := \lim_{m \rightarrow \infty} \theta^{k+(p-1)p^m}.$$

To see that this is well-defined, see [19, Section 4.5]. For $k > 0$ and $f(t) \in \mathcal{W}[[t-1]]$, it is well known (for example, via [17, 3.5(5)]) that

$$\theta^k f(t) = \int_{\mathbb{Z}_p^\times} t^x x^k df,$$

and the same identity also holds for $k < 0$. Thus we may re-write the definition of $\mathcal{L}_\mathbf{p}(f)$ as

$$\begin{aligned} \mathcal{L}_\mathbf{p}(f)(\rho) &= \sum_{[\mathbf{a}] \in \text{Pic}\mathcal{O}_K} N(\mathbf{a})^{-r} \cdot (\theta^{-r} \hat{f}_{\mathbf{a}}^b \otimes \rho[\mathbf{a}])(A_{\mathbf{a}}, \eta_{\mathbf{a}}) \\ &= (\sqrt{-D_K})^r \sum_{[\mathbf{a}] \in \text{Pic}\mathcal{O}_K} ((\theta^{-r} \hat{f}^b)_{\mathbf{a}} \otimes \rho[\mathbf{a}])(A_{\mathbf{a}}, \eta_{\mathbf{a}}) \end{aligned}$$

for any continuous function $\rho : \tilde{\Gamma} \rightarrow \mathcal{O}_{\mathbb{C}_p}$.

5 Congruent modular forms

As before, we denote by $f_1 \in S_{2r_1}^{\text{new}}(\Gamma_0(N_1))$, $f_2 \in S_{2r_2}(\Gamma_0(N_2))^{\text{new}}$ normalized Hecke eigenforms of weight $2r_1$, $2r_2$ and levels N_1, N_2 , respectively. Suppose that both f_1 and f_2 satisfy Hypothesis (Heeg). Then there exist ideals $\mathfrak{N}_1, \mathfrak{N}_2$ in $\mathcal{O}_K$ such that $N_1\mathcal{O}_K = \mathfrak{N}_1\overline{\mathfrak{N}_1}$ and $N_2\mathcal{O}_K = \mathfrak{N}_2\overline{\mathfrak{N}_2}$. Further assume that for every $\ell \mid \gcd(N_1, N_2)$, one has $\gcd(\ell, \mathfrak{N}_1) = \gcd(\ell, \mathfrak{N}_2)$ so that $\mathcal{O}_K/\text{lcm}(\mathfrak{N}_1, \mathfrak{N}_2) \simeq \mathbb{Z}/\text{lcm}(N_1, N_2)\mathbb{Z}$.

We first show that $\mathcal{L}_\mathbf{p}(f_1)$ and $\mathcal{L}_\mathbf{p}(f_2)$ are congruent when their q -expansions are congruent.

Lemma 5.1. *Suppose that $f_1 \in S_{2r_1}(\Gamma_0(N_1))^{\text{new}}$, $f_2 \in S_{2r_2}^{\text{new}}(\Gamma_0(N_2))$ have the same level $N_1 = N_2$. Let L be a number field containing $\mathbb{Q}(\{a_n(f_1), a_n(f_2)\}_{n>0})$ and let ϖ be a uniformizer of $\mathcal{O}_{L_p}$. Suppose that $a_n(f_1) \equiv a_n(f_2) \pmod{\varpi^m \mathcal{O}_{L_p}}$ for every*

n . Then we have the congruences $f_1^\flat \equiv f_2^\flat \pmod{\varpi^m \mathcal{O}_{L_p}}$, $\widehat{f}_1^\flat \equiv \widehat{f}_2^\flat \pmod{\varpi^m \mathcal{O}_{L_p}}$ between p -adic modular forms, and $\mathcal{L}_p(f_1) \equiv \mathcal{L}_p(f_2) \pmod{\varpi^m \mathcal{W}[\Gamma_K^-]}$.

Proof. The congruences between p -adic modular forms follow from the q -expansion principle [12, Corollary 3.5]. We show that $\mathcal{L}_p(f_1)(\rho) \equiv \mathcal{L}_p(f_2)(\rho) \pmod{\varpi^m \mathcal{O}_{\mathbb{C}_p}}$ for every continuous map $\rho : \widetilde{\Gamma} \rightarrow \mathcal{O}_{\mathbb{C}_p}^\times$, and the congruence $\mathcal{L}_p(f_1) \equiv \mathcal{L}_p(f_2) \pmod{\varpi^m \mathcal{W}[\Gamma_K^-]}$ follows by the same argument as [20, Theorem (1.10)]. Let $\chi_{\text{cyc}} : G_{\mathbb{Q}} \rightarrow \mathbb{Z}_p^\times \subset \mathcal{O}_{L_p}^\times$ be the cyclotomic character and let ρ_{f_i} be the Weil-Deligne representation attached to f_i for $i \in \{1, 2\}$. Since $\det(\rho_{f_i}) = \chi_{\text{cyc}}^{2r_i-1}$, we have the congruence

$$\chi_{\text{cyc}}^{2r_1-1} \equiv \chi_{\text{cyc}}^{2r_2-1} \pmod{1 + \varpi^m \mathcal{O}_{L_p}}.$$

Suppose that $\varpi^m \mathcal{O}_{L_p} \cap \mathbb{Z}_p = p^{m'} \mathbb{Z}_p$, then the congruence above actually holds in $(\mathbb{Z}_p/p^{m'} \mathbb{Z}_p)^\times \subset (\mathcal{O}_{L_p}/\varpi^m \mathcal{O}_{L_p})^\times$:

$$\chi_{\text{cyc}}^{2r_1-1} \equiv \chi_{\text{cyc}}^{2r_2-1} \pmod{1 + p^{m'} \mathbb{Z}_p}.$$

Hence, we have the congruence $2r_1 \equiv 2r_2 \pmod{\varphi(p^{m'})}$.

Given a continuous function $\rho : \widetilde{\Gamma} \rightarrow \mathcal{O}_{\mathbb{C}_p}$, we may write

$$\mathcal{L}_p(f_1)(\rho) = \sum_{[\mathfrak{a}] \in \text{Pic} \mathcal{O}_K} N(\mathfrak{a})^{-r_1} \cdot (\theta^{-r_1} \widehat{f}_{1,\mathfrak{a}}^\flat \otimes \rho|[\mathfrak{a}])(A_{\mathfrak{a}}, \eta_{\mathfrak{a}})$$

$$\mathcal{L}_p(f_2)(\rho) = \sum_{[\mathfrak{a}] \in \text{Pic} \mathcal{O}_K} N(\mathfrak{a})^{-r_2} \cdot (\theta^{-r_2} \widehat{f}_{2,\mathfrak{a}}^\flat \otimes \rho|[\mathfrak{a}])(A_{\mathfrak{a}}, \eta_{\mathfrak{a}})$$

If $r_1 \equiv r_2 \pmod{\phi(p^{m'})}$, then $n^{r_1} \equiv n^{r_2} \pmod{p^{m'}}$ for every $n \in \mathbb{Z}_p^\times$ and the result follows immediately. Otherwise, $n^{r_1} \equiv \left(\frac{n}{p}\right) n^{r_2} \pmod{p^{m'}}$ where $\left(\frac{\cdot}{p}\right)$ is the Legendre symbol on $\mathbb{F}_p^\times$ defined as $\left(\frac{x}{p}\right) = x^{(p-1)/2}$. With a slight abuse of notation, we will also denote by $\left(\frac{\cdot}{p}\right)$ the lift of the Legendre symbol to $\mathbb{Z}_p^\times$. Since $\left(\frac{\cdot}{p}\right) \otimes t^m = \left(\frac{m}{p}\right) t^m$ [17, 85] and $n^{r_1} \equiv \left(\frac{n}{p}\right) n^{r_2} \pmod{p^{m'}}$, we have the congruence

$$\theta^{-r_1} \widehat{f}_{1,\mathfrak{a}}^\flat(t) \equiv \left(\frac{\cdot}{p}\right) \otimes \theta^{-r_2} \widehat{f}_{2,\mathfrak{a}}^\flat(t) \pmod{\varpi^m \mathcal{W}[\mathbb{T} - 1]}.$$

Moreover, one also has $N(\mathfrak{a})^{-r_1} \equiv \left(\frac{N(\mathfrak{a})}{p}\right) N(\mathfrak{a})^{-r_2}$, from which it follows that

$$\mathcal{L}_p(f_1)(\rho) \equiv \sum_{[\mathfrak{a}] \in \text{Pic} \mathcal{O}_K} N(\mathfrak{a})^{-r_2} \left(\frac{N(\mathfrak{a})}{p}\right) \rho(\mathfrak{a}) \left(\theta^{-r_2} \widehat{f}_2^\flat \otimes \left(\frac{\cdot}{p}\right) \rho_p \right) (A_{\mathfrak{a}}, \eta_{\mathfrak{a}}) \pmod{\varpi^m \mathcal{W}}. \quad (5.1)$$

We define ψ to be the Hecke character such that $\psi(\mathfrak{a}) = \left(\frac{N(\mathfrak{a})}{p}\right)$ for prime-to- p fractional ideals $\mathfrak{a}$ of K . Then ψ is an anticyclotomic Hecke character of order 2 and

conductor p , and $\psi_p : \mathcal{O}_{K,p}^\times \rightarrow \{\pm 1\}$ is the Legendre symbol $\left(\frac{\cdot}{p}\right)$. We may now rewrite the congruence (5.1) as

$$\mathcal{L}_p(f_1)(\rho) \equiv \sum_{[\mathfrak{a}] \in \text{Pic } \mathcal{O}_K} N(\mathfrak{a})^{-r_2} (\psi\phi)(\mathfrak{a}) (\theta^{-r_2} \widehat{f}_2^b \otimes \psi_p \rho_p)(A_{\mathfrak{a}}, \eta_{\mathfrak{a}}) \pmod{\varpi^m \mathcal{W}}.$$

In other words,

$$\mathcal{L}_p(f_1) \equiv \text{Tw}_\psi \mathcal{L}_p(f_2) \pmod{\varpi^m \mathcal{W}[\widetilde{\Gamma}]}. \quad (5.2)$$

Since ψ is a Hecke character of order 2 and p is odd, the restriction of ψ to the anticyclotomic $\mathbb{Z}_p$ -extension Γ_K^- is trivial. Hence, one has the congruence

$$\mathcal{L}_p(f_1) \equiv \mathcal{L}_p(f_2) \pmod{\varpi^m \mathcal{W}[\Gamma_K^-]}.$$

□

5.1 Hecke operators at p in Serre-Tate coordinates

We recall some Hecke operators in terms of the complex uniformization of Igusa schemes. Let $\mathfrak{a}$ be a fractional ideal of $\mathcal{O}_K$ and let $x_{\mathfrak{a}} = [(A_{\mathfrak{a}}, \eta_{\mathfrak{a}})] = [\vartheta, \rho(\bar{a})^{-1} \zeta_c]$ (see Section 3). For $z \in \mathbb{Q}_p$, we define $\mathfrak{n}(z) := \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} \in \text{GL}_2(\mathbb{Q}_p) \subset \text{GL}_2(\widehat{\mathbb{Q}})$ and let $x_{\mathfrak{a}} * \mathfrak{n}(z) := [\vartheta, \rho(\bar{a})^{-1} \zeta_c \mathfrak{n}(z)]$ under the action of $\text{GL}_2(\widehat{\mathbb{Q}})$ on $\text{Ig}(N)(\mathbb{C})$.

By [5, Proposition 3.3], for a primitive Dirichlet character $\phi : (\mathbb{Z}/p^n \mathbb{Z})^\times \rightarrow \mathcal{O}_{\mathbb{C}_p}^\times$, the integral in Definition 4.1 can be written as

$$f_{\mathfrak{a}} \otimes \phi(x_{\mathfrak{a}}) = p^{-n} \mathfrak{g}(\phi) \sum_{u \in (\mathbb{Z}/p^n \mathbb{Z})^\times} \phi^{-1}(u) f(x_{\mathfrak{a}} * \mathfrak{n}(up^{-n})).$$

In [4, Proposition 6.4], the author discusses the moduli interpretations of $x_{\mathfrak{a}} * \mathfrak{n}(up^{-n})$ for $u \in (\mathbb{Z}/p^n \mathbb{Z})^\times$ as quotients of $A_{\mathfrak{a}}$ by certain rank- p^n subgroup schemes of $A_{\mathfrak{a}}[p^\infty]$. Moreover we have $x_{\mathfrak{a}} * \mathfrak{n}(up^{-n}) \otimes \widehat{\mathbb{F}}_p = x_{\mathfrak{a}} \otimes \widehat{\mathbb{F}}_p$, and the Serre-Tate coordinate of $x_{\mathfrak{a}} * \mathfrak{n}(up^{-n})$ is given by $t_{\mathfrak{a}}(x_{\mathfrak{a}} * \mathfrak{n}(up^{-n})) = \zeta_{p^n}^{-u \mathbb{N}(\mathfrak{a})^{-1} \sqrt{-D_K}^{-1}}$ [5, Lemma 3.2].

5.2 Hecke operators at $\ell \neq p$ in Serre-Tate coordinates

Let $f \in S_{2r}(\Gamma_0(N))^{\text{new}}$ be a normalized newform of weight $2r$ and level N that is an eigenform for all Hecke operators.

For $\gcd(N, \ell) \mid N^\sharp$, we may naturally identify f as a form of level $N^\sharp$. For an ordinary test triplet $(A, \eta_{N^\sharp}, \omega) \in \mathcal{M}_{\Gamma_1(N^\sharp)}$ with level $N^\sharp$ structure $\eta_{N^\sharp}$, let $C \subset A[N^\sharp]$ be the image of the level structure $\eta_{N^\sharp}$.

Let π be the projection $A \rightarrow A/C[\ell]$. Note that the morphism

$$\pi \circ \eta_{N^\sharp} : \mu_{N^\sharp} \rightarrow C/C[\ell]$$

has kernel μ_ℓ , and we will denote by $\overline{\pi \circ \eta_{N^\sharp}}$ the isomorphism

$$\overline{\pi \circ \eta_{N^\sharp}} : \mu_{N^\sharp} / \mu_\ell \rightarrow C/C[\ell].$$

Moreover, we will denote by $(\cdot)^{1/\ell}$ the inverse of the isomorphism $\mu_{N^\sharp} / \mu_\ell \xrightarrow{\zeta \rightarrow \zeta^\ell} \mu_{N^\sharp \ell^{-1}}$. One can define the 'dividing by ℓ -level structure' operator V_ℓ on ordinary test triplets as

$$V_\ell(A, \eta_{N^\sharp}, \omega) = (A/C[\ell], \pi \circ \eta_{N^\sharp} \circ (\cdot)^{1/\ell}, \tilde{\pi}^* \omega),$$

where $\pi : A \rightarrow A/C[\ell]$ is the canonical projection and $\tilde{\pi} : A/C[\ell] \rightarrow A$ is its dual isogeny.

This induces an operator V_ℓ^* on the space of classical modular forms of level $\Gamma_0(N^\sharp)$ via the rule $V_\ell^* f(A, \eta_{N^\sharp}, \omega) = f(V_\ell(A, \eta_{N^\sharp}, \omega))$, which acts on q -expansions as $f(q) \mapsto f(q^\ell)$ [10, 14, 15].

Definition 5.2. We define the (ℓ) -stabilization for a newform f of conductor N and weight $2r$ as:

$$f^{(\ell)} = \begin{cases} f - a_\ell(f) V_\ell^* f + \ell^{2r-1} V_\ell^* V_\ell^* f & \text{if } \ell \nmid N, \\ f - a_\ell(f) V_\ell^* & \text{otherwise.} \end{cases}$$

where we implicitly identify f as a form of level $N^\sharp$.

We now give a description of the Hecke operators above for a p -adic modular form $f \in V_p(N, \mathcal{W})$ of level N . Suppose that $N \mid N^\sharp$ and let $(A, \eta_{N^\sharp} \times \eta_p) \in \text{Ig}(N^\sharp)$. There is a natural map

$$\begin{aligned} \frac{N^\sharp}{N} : \text{Ig}(N^\sharp) &\rightarrow \text{Ig}(N) \\ (A, \eta_{N^\sharp} \times \eta_p) &\mapsto (A, \eta_N \times \eta_p), \end{aligned}$$

where η_N is the restriction of $\eta_{N^\sharp}$ to μ_N .

This induces an identification of p -adic modular forms of level N as forms of level $N^\sharp$:

$$[N^\sharp/N]^* : V_p(N, \mathcal{W}) \hookrightarrow V_p(N^\sharp, \mathcal{W}).$$

For $\gcd(N, \ell) \mid N^\sharp$, we define the following analogue of the V_ℓ operator for p -adic modular forms:

$$\begin{aligned} V_\ell : \text{Ig}(N^\sharp) &\rightarrow \text{Ig}(N^\sharp \ell^{-1}) \\ (A, \eta) &\mapsto (A/C[\ell], \overline{\pi \circ \eta_{N^\sharp}} \circ (\cdot)^{1/\ell} \times \tilde{\pi}^{-1} \circ \eta_p) \end{aligned}$$

for $C[\ell] := \text{im}(\eta)$, and similarly define $V_\ell^* f(A, \eta) = f(V_\ell(A, \eta))$ for a p -adic modular form f of level $N^\sharp$. We also note that $\tilde{\pi}^{-1} \circ \eta_p = \frac{1}{\ell} \circ \pi \circ \eta_p$.

For a complete local $\mathcal{W}$ -algebra R and $[(A, \eta)_R] \in \text{Ig}(N)(R)$, recall from Section 2 that the $\Gamma_1(Np^\infty)$ -level structure $\eta = \eta_N \oplus \eta_p$ determines a map $\hat{\eta}_p : \hat{\mathbf{G}}_m \xrightarrow{\sim} \hat{A}$ [14, (1.10.1)] (see also [15, Proposition 1]), which defines a differential $\omega(\hat{\eta}_p) : \text{Lie}(\hat{A}) \simeq \text{Lie}(\hat{\mathbf{G}}_m) \rightarrow \text{Lie}(\hat{\mathbf{G}}_m) = R$. A geometric modular form f can then be identified as a p -adic modular form via the rule $\hat{f}(A, \eta) = f(A, \eta, \omega(\hat{\eta}_p))$. To show the compatibility of the V_ℓ operator defined on geometric modular forms and p -adic modular forms, we begin with the following

Lemma 5.3. *Let $\phi : A/R \rightarrow A'/R$ be an isogeny of elliptic curves. Suppose that η_p is a p^∞ -level structure on A/R , and $\phi \circ \eta_p$ is the p^∞ -level structure on A'/R induced by ϕ . Then $(\phi^*)^{-1}\omega(\widehat{\eta_p}) = \omega(\widehat{\phi \circ \eta_p})$, where $\phi^* : H^0(A'/R, \underline{\Omega}_{A'/R}^1) \rightarrow H^0(A/R, \underline{\Omega}_{A/R}^1)$ is induced by ϕ .*

Proof. Throughout this proof, we use the equivalence between the category of divisible commutative Lie groups and the category of connected p -divisible groups [15, Proposition 1].

Let $\phi : A/R \rightarrow A'/R$ be an isogeny. Then there are induced maps

$$\widehat{\mathbf{G}}_m \xrightarrow{\widehat{\eta_p}} \widehat{A} \xrightarrow{\widehat{\phi}} \widehat{A'},$$

$$\mathrm{Lie}(\widehat{\mathbf{G}}_m) \xrightarrow{\mathrm{Lie}(\widehat{\eta_p})} \mathrm{Lie}(\widehat{A}) \xrightarrow{\mathrm{Lie}(\widehat{\phi})} \mathrm{Lie}(\widehat{A'}).$$

Recall that $\omega(\widehat{\eta_p})$ (respectively $\omega(\widehat{\phi \circ \eta_p})$) is defined as the inverse of $\mathrm{Lie}(\widehat{\eta_p})$ (respectively $\mathrm{Lie}(\widehat{\phi \circ \eta_p})$):

$$\omega(\widehat{\eta_p}) : \mathrm{Lie}(\widehat{A}) \simeq \mathrm{Lie}(\widehat{\mathbf{G}}_m) \xrightarrow{\mathrm{Lie}(\widehat{\eta_p})^{-1}} \mathrm{Lie}(\widehat{\mathbf{G}}_m) = R,$$

$$\omega(\widehat{\phi \circ \eta_p}) : \mathrm{Lie}(\widehat{A'}) \simeq \mathrm{Lie}(\widehat{\mathbf{G}}_m) \xrightarrow{\mathrm{Lie}(\widehat{\phi \circ \eta_p})^{-1}} \mathrm{Lie}(\widehat{\mathbf{G}}_m) = R.$$

Hence, we have $\mathrm{Lie}(\widehat{\phi})^* \omega(\widehat{\phi \circ \eta_p}) = \omega(\widehat{\eta_p})$ by functoriality, where $\mathrm{Lie}(\widehat{\phi})^*$ is the pull-back map induced by $\mathrm{Lie}(\widehat{\phi})$. Moreover, the map $\mathrm{Lie}(\widehat{\phi})^*$ is the same as $\phi^* : H^0(A'/R, \underline{\Omega}_{A'/R}^1) \rightarrow H^0(A/R, \underline{\Omega}_{A/R}^1)$, and we have $\phi^* \omega(\widehat{\phi \circ \eta_p}) = \omega(\widehat{\eta_p})$. $\square$

Lemma 5.4. *Let $\widehat{f} \in V_p(N, \mathcal{W})$ be the p -adic avatar of a geometric modular form f . Then*

$$V_\ell^* \widehat{f} = \widehat{V_\ell^* f},$$

where the V_ℓ^* operator on the left-hand side acts on p -adic modular forms and the V_ℓ^* operator on the right-hand side acts on geometric modular forms.

Proof. This follows directly from $\tilde{\pi}^* \omega(\widehat{\eta_p}) = \omega(\widehat{\tilde{\pi}^{-1} \circ \eta_p})$ by the previous lemma, and the definitions

$$\widehat{V_\ell^* f}(A, \eta) = f(A/C[\ell], \overline{\pi \circ \eta_{N^\#}} \circ (\cdot)^{1/\ell}, \tilde{\pi}^* \omega(\eta_p)),$$

$$V_\ell^* \widehat{f}(A, \eta) = f(A/C[\ell], \overline{\pi \circ \eta_{N^\#}} \circ (\cdot)^{1/\ell}, \omega(\tilde{\pi}^{-1} \circ \eta_p)).$$

$\square$

Let ℓ be a prime that splits in $\mathcal{O}_K$ as $\ell = v\bar{v}$, and let $N^\#$ be such that $\mathrm{lcm}(N, \ell^2) \mid N^\#$. For every fractional ideal $\mathfrak{a}$ of $\mathcal{O}_K$ and every level M divisible by N , let $x_{\mathfrak{a}} = (A_{\mathfrak{a}}, \eta_{\mathfrak{a}}) \in \mathrm{Ig}(M)$ be a CM point satisfying $\mathrm{im}(\eta_{\mathfrak{a}})[\ell^\infty] = \mathrm{im}(\eta_{\mathfrak{a}})[\ell^\infty] \cap A[v^\infty]$. We assume that these points are compatible with the projections $\mathrm{Ig}(M') \rightarrow \mathrm{Ig}(M)$ for $M \mid M'$. It follows from definitions that the value of a p -adic modular form $f \in V_p(N, \mathcal{W})$ at such a CM point does not depend on the implicit level under the natural identification $V_p(N, \mathcal{W}) \hookrightarrow V_p(M, \mathcal{W})$ for any M divisible by N .

Lemma 5.5. *Let $x_{\mathbf{a}} = (A_{\mathbf{a}}, \eta_{\mathbf{a}}) \in Ig(N^{\sharp})$ be a CM point. Then $V_{\ell}(x_{\mathbf{a}}) = (A_{\bar{v}^{-1}\mathbf{a}}, \eta_{\bar{v}^{-1}\mathbf{a}}) \in Ig(N^{\sharp}\ell^{-1})$. As a consequence, we have*

$$V_{\ell}^* f(x_{\mathbf{a}}) = f(x_{\bar{v}^{-1}\mathbf{a}})$$

for a p -adic modular form $f \in V_p(N, \mathcal{W})$.

Proof. For ease of notation, we will denote $A = A_{\mathbf{a}}$ and $\eta = \eta_{\mathbf{a}}$. We denote by π_v the projection $A \rightarrow A/A[v]$, and by $\pi_{\bar{v}}$ the projection $A_v \rightarrow A_v/A_v[\bar{v}]$ where $A_v = A/A[v]$. Observe that

$$\begin{aligned} \bar{v} \star V_{\ell}(x_{\mathbf{a}}) &= \bar{v} \star (A/A[v], \overline{\pi_v \circ \eta_{N^{\sharp}}} \circ (\cdot)^{1/\ell} \times \tilde{\pi}_v^{-1} \circ \eta_p) \\ &= (A/A[\ell], \pi_{\bar{v}} \circ \overline{\pi_v \circ \eta_{N^{\sharp}}} \circ (\cdot)^{1/\ell} \times \pi_{\bar{v}} \circ \tilde{\pi}_v^{-1} \circ \eta_p). \end{aligned}$$

We claim that the isomorphism

$$\iota : A/A[\ell] \rightarrow A$$

$$x + A[\ell] \mapsto [\ell]x$$

introduced in [10, Lemma 3.5] gives rise to the isomorphism between the tuples

$$(A/A[\ell], \pi_{\bar{v}} \circ \overline{\pi_v \circ \eta_{N^{\sharp}}} \circ (\cdot)^{1/\ell} \times \pi_{\bar{v}} \circ \tilde{\pi}_v^{-1} \circ \eta_p) \simeq (A, (\eta_{N^{\sharp}/\ell} \times \eta_p)),$$

where $\eta_{N^{\sharp}/\ell-1}$ is the restriction of $\eta_{N^{\sharp}}$ to $\mu_{N^{\sharp}/\ell-1}$.

Indeed, following the argument of [10, Lemma 3.5], the composition $\iota \circ \pi_{\bar{v}} \circ \pi_v$ is the multiplication by ℓ map $[\ell] : A \rightarrow A$. This implies that the dual isogeny $\tilde{\pi}_v$ of π_v is $\iota \circ \pi_{\bar{v}}$, so that $\iota \circ \pi_{\bar{v}} \circ \tilde{\pi}_v^{-1} \circ \eta_p = \eta_p$.

Next, we show that $\iota \circ \pi_{\bar{v}} \circ \overline{\pi_v \circ \eta_{N^{\sharp}}} \circ (\cdot)^{1/\ell} = \eta_{N^{\sharp}/\ell-1}$. Since $\iota \circ \pi_{\bar{v}} \circ \pi_v$ is just the multiplication by ℓ map, the composition $\iota \circ \pi_{\bar{v}} \circ \pi_v \circ \eta_{N^{\sharp}}$ is simply

$$\mu_{N^{\sharp}} \xrightarrow{\eta_{N^{\sharp}}} A[\mathfrak{N}^{\sharp}] \xrightarrow{[\ell]} A[\mathfrak{N}^{\sharp}v^{-1}].$$

The following diagram commutes:

$$\begin{array}{ccc} \mu_{N^{\sharp}} & \xrightarrow{\eta_{N^{\sharp}}} & A[\mathfrak{N}^{\sharp}] \\ \downarrow (\cdot)^{\ell} & & \downarrow \ell \\ \mu_{N^{\sharp}/\ell-1} & \xrightarrow{\eta_{N^{\sharp}/\ell-1}} & A[\mathfrak{N}^{\sharp}v^{-1}], \end{array}$$

which shows that $\iota \circ \pi_{\bar{v}} \circ \overline{\pi_v \circ \eta_{N^{\sharp}}} \circ (\cdot)^{1/\ell} = \eta_{N^{\sharp}/\ell-1}$. $\square$

If $x \otimes \bar{\mathbb{F}}_p = x_{\mathbf{a}} \otimes \bar{\mathbb{F}}_p$, then the reduction $V_{\ell}(x) \otimes \bar{\mathbb{F}}_p$ of $V_{\ell}(x)$ is $x_{\bar{v}^{-1}\mathbf{a}} \otimes \bar{\mathbb{F}}_p$. Analogous to [19, Lemma 4.8], the relationship between their t -expansions is given by:

$$t_{\bar{v}^{-1}\mathbf{a}}(V_{\ell}(x)) = t_{\mathbf{a}}(x)^{\ell}.$$

It also follows from this identity that $V_\ell(x_{\mathfrak{a}} * \mathfrak{n}(up^{-n})) = x_{\bar{v}^{-1}\mathfrak{a}} * \mathfrak{n}(up^{-n})$. Indeed,

$$\begin{aligned} t_{\bar{v}^{-1}\mathfrak{a}}(V_\ell(x_{\mathfrak{a}} * \mathfrak{n}(up^{-n}))) &= t_{\mathfrak{a}}(x_{\mathfrak{a}} * \mathfrak{n}(up^{-n}))^\ell \\ &= \zeta_{p^n}^{-u\ell\mathbb{N}(\mathfrak{a})^{-1}\sqrt{-D_K}^{-1}} \\ &= \zeta_{p^n}^{-u\mathbb{N}(\bar{v}^{-1}\mathfrak{a})^{-1}\sqrt{-D_K}^{-1}} \\ &= t_{\bar{v}^{-1}\mathfrak{a}}(x_{\bar{v}^{-1}\mathfrak{a}} * \mathfrak{n}(up^{-n})). \end{aligned}$$

Lemma 5.6. *Let $\phi: \mathbb{Z}_p^\times \rightarrow \mathcal{O}_{\mathbb{C}_p^\times}$ be a p -adic character of conductor p^n . We have the following identity: $((\theta^{-r}V_\ell^*f)_{\mathfrak{a}} \otimes \phi)(x_{\mathfrak{a}}) = \ell^{-r}((\theta^{-r}f)_{\bar{v}^{-1}\mathfrak{a}} \otimes \phi)(x_{\bar{v}^{-1}\mathfrak{a}})$.*

Proof. By examining t -expansions, observe that

$$\theta^{-r}V_\ell^*f = \ell^{-r}V_\ell^*\theta^{-r}f$$

for a p -adic modular form $f \in V_p(N, \mathcal{W})$. Combined with Lemma 5.5, we have:

$$\begin{aligned} ((\theta^{-r}V_\ell^*f)_{\mathfrak{a}} \otimes \phi)(x_{\mathfrak{a}}) &= p^{-n}\mathfrak{g}(\phi) \cdot \sum_{u \in (\mathbb{Z}/p^n\mathbb{Z})^\times} \phi^{-1}(u)(\theta^{-r}V_\ell^*f)(x_{\mathfrak{a}} * \mathfrak{n}(up^{-n})) \\ &= p^{-n}\mathfrak{g}(\phi)\ell^{-r} \cdot \sum_{u \in (\mathbb{Z}/p^n\mathbb{Z})^\times} \phi^{-1}(u)(V_\ell^*\theta^{-r}f)(x_{\mathfrak{a}} * \mathfrak{n}(up^{-n})) \\ &= p^{-n}\mathfrak{g}(\phi)\ell^{-r} \cdot \sum_{u \in (\mathbb{Z}/p^n\mathbb{Z})^\times} \phi^{-1}(u)(\theta^{-r}f)(x_{\bar{v}^{-1}\mathfrak{a}} * \mathfrak{n}(up^{-n})) \\ &= \ell^{-r}((\theta^{-r}f)_{\bar{v}^{-1}\mathfrak{a}} \otimes \phi)(x_{\bar{v}^{-1}\mathfrak{a}}). \end{aligned}$$

□

Definition 5.7. Following [1, 8, 9], we define $\mathcal{P}_v \in \mathcal{W}[\Gamma_K^-]$ such that

$$\mathcal{P}_v(f) = \begin{cases} 1 - a_\ell(f)\ell^{-r} \cdot \gamma_v + \ell^{-1} \cdot \gamma_v^2 \in \mathcal{W}[\Gamma_K^-] & \text{if } \ell \nmid N, \\ 1 - a_\ell(f)\ell^{-r} \cdot \gamma_v & \text{if } \ell \mid N. \end{cases}$$

where $\gamma_v \in \Gamma_K^-$ is the Frobenius at v . We define $\mathcal{P}_{\bar{v}}(f) \in \mathcal{W}[\tilde{\Gamma}]$ similarly.

We fix a topological generator γ_0 of Γ_K^- , and let $\mathcal{W}[\Gamma_K^-] \simeq \mathcal{W}[T]$ be the isomorphism given by $\gamma_0 \mapsto T + 1$.

Lemma 5.8. *As elements in $\mathcal{W}[\Gamma_K^-]$, both $\mathcal{P}_v(f)$ and $\mathcal{P}_{\bar{v}}(f)$ have μ -invariants 0.*

Proof. One may write $\gamma_v = \gamma_0^a$ where $a \in \mathbb{Z}_p$. For $\ell \mid N$, $\mathcal{P}_{\bar{v}} = 1 - a_\ell(f)\ell^{-r} \cdot (1 + T)^a$. Let $a = \sum_{n \geq k} a_n p^n$, where $a_n \in \{0, \dots, p-1\}$ and k is the smallest index such that $a_k \neq 0$. One has the following congruence:

$$(1 + T)^{f_v} \equiv \prod_{n \geq k} (1 + T^{p^n})^{a_n} \pmod{\varpi},$$

from which it follows that

$$\mathcal{P}_v(f)(T) \equiv 1 - a_\ell(f)\ell^{-r}(1 + T^{p^k})^{a_k} \equiv 1 - a_\ell(f)\ell^{-r}(1 + a_k T^{p^k}) \pmod{(\varpi, T^{2p^k})},$$

and therefore $\mathcal{P}_v(f)(T) \not\equiv 0 \pmod{\varpi}$. The analogous statement for $\mathcal{P}_{\bar{v}}(f)$ also holds.

We can similarly show that $\mu(\mathcal{P}_\ell(f)) = 0$ for $\ell \nmid N$. Indeed, we may write $\mathcal{P}_v(f) = (1 - a_\ell \cdot \gamma_v)(1 - b_\ell \cdot \gamma_v)$, and it can be shown by the same argument as above that both $1 - a_\ell \cdot \gamma_v, 1 - b_\ell \cdot \gamma_v$ have μ -invariants 0. The same argument applies to $\mathcal{P}_{\bar{v}}(f)$. $\square$

Theorem 5.9. *Let $f^{(\ell)}$ be the ℓ -depletion of f , considered as a geometric modular form of level $N^\sharp$ where $\text{lcm}(\ell, N) \mid N^\sharp$. Then $\mathcal{L}_\mathbf{p}(f^{(\ell)}) = \mathcal{P}_{\bar{v}}(f)\mathcal{L}_\mathbf{p}(f)$.*

Proof. For every locally algebraic character $\rho \in \mathfrak{X}_{p^\infty}$, we use Lemma 5.6 to obtain the following:

$$\begin{aligned} \mathcal{L}_\mathbf{p}(V_\ell^* f)(\rho) &= (\sqrt{-D_K})^r \sum_{[\mathbf{a}] \in \text{Pic } \mathcal{O}_K} ((\theta^{-r} V_\ell^* \hat{f}^\flat)_\mathbf{a} \otimes \rho|[\mathbf{a}])(A_\mathbf{a}, \eta_\mathbf{a}) \\ &= \rho(\bar{v})\ell^{-r}(\sqrt{-D_K})^r \sum_{[\mathbf{a}] \in \text{Pic } \mathcal{O}_K} ((\theta^{-r} \hat{f}^\flat)_{\bar{v}^{-1}\mathbf{a}} \otimes \rho|[\bar{v}^{-1}\mathbf{a}])(A_{\bar{v}^{-1}\mathbf{a}}, \eta_{\bar{v}^{-1}\mathbf{a}}) \\ &= \rho(\bar{v})\ell^{-r} \mathcal{L}_\mathbf{p}(f)(\rho). \end{aligned}$$

Hence $\mathcal{L}_\mathbf{p}(f^{(\ell)}) = \mathcal{L}_\mathbf{p}(f - a_\ell(f)V_\ell^* f)(\rho) = (1 - a_\ell(f)\rho(\bar{v})\ell^{-r})\mathcal{L}_\mathbf{p}(f)$ for $\ell \mid N$, and for $\ell \nmid N$ we have $\mathcal{L}_\mathbf{p}(f^{(\ell)}) = (1 - a_\ell(f)\rho(\bar{v})\ell^{-r} + \rho(\bar{v})^2\ell^{-1})\mathcal{L}_\mathbf{p}(f)$. $\square$

Theorem 5.10. *Suppose that $f_1 \in S_{2r_1}(\Gamma_0(N_1))^{new}$, $f_2 \in S_{2r_2}(\Gamma_0(N_2))^{new}$ are new-forms satisfying Hypothesis (Heeg) whose coefficients lie in some number field L . We assume that $\mathcal{W}$ is the ring of integers of a finite extension of $\mathbb{Q}_p^{nr}$ containing L .*

Suppose that the induced semi-simplified mod ϖ^m Galois representations: $\bar{\rho}_{f_1}, \bar{\rho}_{f_2} : \text{Gal}(\bar{L}/L) \rightarrow GL_2(\mathcal{O}_{L_p}/\varpi^m \mathcal{O}_{L_p})$ are isomorphic, where ϖ is the uniformizer of $\mathcal{O}_{L_p}$. For each prime $\ell \mid N_1 N_2$, let $v \mid \mathfrak{N}_1 \mathfrak{N}_2$ be the corresponding prime above ℓ . Then the following congruence holds:

$$\prod_{\ell \mid N_1 N_2} \mathcal{P}_{\bar{v}}(f_1)\mathcal{L}_\mathbf{p}(f_1) \equiv \prod_{\ell \mid N_1 N_2} \mathcal{P}_{\bar{v}}(f_2)\mathcal{L}_\mathbf{p}(f_2) \pmod{\varpi^m \mathcal{W}[\Gamma_K^-]}.$$

Moreover, one has the following:

1. $\mu(\mathcal{L}_\mathbf{p}(f_1)) = 0$ if and only if $\mu(\mathcal{L}_\mathbf{p}(f_2)) = 0$.
2. Assuming that $\mu(\mathcal{L}_\mathbf{p}(f_1)) = \mu(\mathcal{L}_\mathbf{p}(f_2)) = 0$,

$$\sum_{\ell \mid N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_1)) + \lambda(\mathcal{L}_\mathbf{p}(f_1)) = \sum_{\ell \mid N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_2)) + \lambda(\mathcal{L}_\mathbf{p}(f_2)).$$

Proof. Let $N^\sharp := \text{lcm}_{\ell \mid N_1 N_2}(N_1, N_2, \ell^2)$, and let $\mathfrak{N}^\sharp = \text{lcm}_{v \mid \mathfrak{N}_1 \mathfrak{N}_2}(\mathfrak{N}_1, \mathfrak{N}_2, v^2)$. Since $f_1^{(N_1 N_2)} \equiv f_2^{(N_1 N_2)} \pmod{\varpi^m}$, Lemma 5.1 gives the following congruence:

$$\mathcal{L}_\mathbf{p}(f_1^{(N_1 N_2)}) \equiv \mathcal{L}_\mathbf{p}(f_2^{(N_1 N_2)}) \pmod{\varpi^m \mathcal{W}[\Gamma_K^-]}.$$

By repeatedly applying Theorem 5.9, we have

$$\mathcal{L}_{\mathfrak{p}}(f^{(N_1 N_2)}) = \left(\prod_{\ell | N_1 N_2} \mathcal{P}_{\bar{v}}(f) \right) \mathcal{L}_{\mathfrak{p}}(f)$$

for each $f \in \{f_1, f_2\}$. Thus, the previous congruence is equivalent to

$$\left(\prod_{\ell | N_1 N_2} \mathcal{P}_{\bar{v}}(f_1) \right) \mathcal{L}_{\mathfrak{p}}(f_1) \equiv \left(\prod_{\ell | N_1 N_2} \mathcal{P}_{\bar{v}}(f_2) \right) \mathcal{L}_{\mathfrak{p}}(f_2) \pmod{\varpi^m \mathcal{W}[\Gamma_K^-]}$$

This congruence also holds over $\mathcal{W}[\Gamma_K^-]/\varpi \mathcal{W}[\Gamma_K^-] \simeq \bar{\mathbb{F}}_p[\Gamma_K^-] \simeq \bar{\mathbb{F}}_p[[T]]$. Since $\mu(\mathcal{P}_{\ell}(f_1)) = \mu(\mathcal{P}_{\ell}(f_2)) = 0$ by Lemma 5.8, we have $\mu(\mathcal{L}_{\mathfrak{p}}(f_1)) = 0$ if and only if $\mu(\mathcal{L}_{\mathfrak{p}}(f_2)) = 0$.

Note that for $F \in \mathcal{W}[\Gamma_K^-][\simeq \mathcal{W}][T]$ with $\mu(F) = 0$, we have $\lambda(F) = \lambda(\bar{F})$, where $\bar{F} \in \bar{\mathbb{F}}_p[[T]]$ is the reduction of $F \pmod{\varpi}$. When $\mu(\mathcal{L}_{\mathfrak{p}}(f_1)) = \mu(\mathcal{L}_{\mathfrak{p}}(f_2)) = 0$, it follows that

$$\sum_{\ell | N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_1)) + \lambda(\mathcal{L}_{\mathfrak{p}}(f_1)) = \sum_{\ell | N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_2)) + \lambda(\mathcal{L}_{\mathfrak{p}}(f_2)).$$

□

6 Applications to generalized Heegner cycles

6.1 Definitions

In this section, we follow the set-up of [5, Section 4]. As before, let $f \in S_{2r}(\Gamma_0(N))^{\text{new}}$ be a normalized Hecke eigenform of weight $2r$ and level N satisfying hypothesis (Heeg).

We may write $K = \mathbb{Q}(\sqrt{-D_K})$ where D_K is the discriminant of K , and for $r > 1$ assume that either $-D_K > 3$ is odd, or $8 \mid D_K$. Such an assumption guarantees a canonical choice of elliptic curve A with CM by $\mathcal{O}_K$, defined over the real subfield of the Hilbert class field H_K of K [5, Section 4.1].

Recall that $V = V_f(r)$ is the self-dual p -adic Galois representation associated with f . For primes p such that $p \nmid 2(2r-1)!N\phi(N)$, we denote by $z_{f,\chi} \in H_f^1(K, T \otimes \chi)$ the generalized Heegner class attached to (f, χ) as constructed in [5, Section 4.5]. We remark that the construction involves the aforementioned canonical CM elliptic curve A .

We recall the definition of the Bloch-Kato logarithm map. Let $\mathbf{B}_{\text{dR}}, \mathbf{B}_{\text{cris}}$ be Fontaine's rings of p -adic periods [21, Definition 5.15, Definition 6.7], and let $t \in \mathbf{B}_{\text{dR}}$ be Fontaine's p -adic analogue of $2\pi i$ [21, Section 5.2.3].

Denote by $\mathbf{D}_{\text{dR},L}(V)$ the filtered $(L \otimes_{\mathbb{Q}_p} F)$ -module $(\mathbf{B}_{\text{dR}} \otimes V)^{G_L}$ and define $H_f^1(L, V) := \ker(H^1(L, V) \rightarrow H^1(L, \mathbf{B}_{\text{cris}} \otimes V))$ in accordance with [22, (3.7.2)]. The

following exponential map is due to Bloch and Kato [22, Section 3]:

$$\exp : \frac{\mathbf{D}_{\mathrm{dR},L}(V)}{\mathrm{Fil}^0 \mathbf{D}_{\mathrm{dR},L}(V)} \rightarrow H_f^1(L, V).$$

The logarithm map is defined as its inverse:

$$\log : H_f^1(L, V) \rightarrow \frac{\mathbf{D}_{\mathrm{dR},L}(V)}{\mathrm{Fil}^0 \mathbf{D}_{\mathrm{dR},L}(V)} = (\mathrm{Fil}^0 \mathbf{D}_{\mathrm{dR},L}(V^*(1)))^\vee.$$

In the special case where V is the p -adic representation attached to an abelian variety, $H_f^1(L, V)$ is the image of the Kummer map in $H^1(L, V)$ and the Bloch-Kato logarithm is the usual logarithm map (see [22, Example 3.11]).

For any field F containing H_K , there is a decomposition

$$H_{\mathrm{dR}}^1(A/F) = H_{\mathrm{dR}}^0(A/F, \Omega_{A/F}^1) \oplus H_{\mathrm{dR}}^1(A/F, \Omega_{A/F}^0).$$

Recall our fixed choice of Néron differential ω_A , and let $\eta_A \in H_{\mathrm{dR}}^1(A/F, \Omega_{A/F}^0)$ such that $\langle \omega_A, \eta_A \rangle = 1$ under the algebraic deRham cup product.

Let $\omega_A^{r-1+j} \eta_A^{r-1-j}$ be a basis of $\mathbf{D}_{\mathrm{dR},F}(\mathrm{Sym}^{2r-2} H_{\mathrm{ét}}^1(A_{\overline{\mathbb{Q}}}, \mathbb{Q}_p))$ as defined in [3, (1.4.6)].

Let W_{2r-2} be the Kuga-Sato variety of dimension $2r-1$. To our cusp form f , we may attach an element in $\tilde{\omega}_f \in H^{2r-1}(W_{2r-2}/F)$ via [3, (1.1.12)] and [3, Lemma 2.2]. Moreover, we may realize V_f as a quotient of $H_{\mathrm{ét}}^{2r-1}(W_{2r-2}/\overline{\mathbb{Q}}, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} F$ by the work of Scholl [23]. Let $\omega_f \in \mathbf{D}(V_f)$ be the image of $\tilde{\omega}_f \in H_{\mathrm{dR}}^{2r-1}(W_{2r-2}/F)$ under the composition

$$H_{\mathrm{dR}}^{2r-1}(W_{2r-2}/F) \simeq \mathbf{D}_{\mathrm{dR}}(H_{\mathrm{ét}}^{2r-1}(W_{2r-2}/\overline{\mathbb{Q}}, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} F) \rightarrow \mathbf{D}_{\mathrm{dR}}(V_f).$$

6.2 A p -adic Gross-Zagier formula

We recall the following p -adic Gross-Zagier formula [5, Theorem 4.9], with the constant term later corrected in the extension to the quaternionic setting due to Magrone [24, Theorem 6.4]. We remark that Theorem 4.9 of CH18 extends the main result of Bertolini-Darmon-Prasanna (see [3, p.1083], [3, Theorem 5.13]) to characters that are ramified at p .

Theorem 6.1. *Suppose $p = \mathfrak{p}\overline{\mathfrak{p}}$ splits in K and let $f \in S_{2r}(\Gamma_0(N))^{new}$ be a Hecke eigen-newform of weight $2r$. If $\chi = \hat{\phi} \in \hat{\mathfrak{X}}_{p^\infty}$ is the p -adic avatar of an anticyclotomic Hecke character of infinity type $(j, -j)$ with $-r < j < r$ and conductor $p^n \mathcal{O}_K$ with $n \geq 1$, then*

$$\frac{\mathcal{L}_{\mathfrak{p}}(f)(\chi^{-1})}{\Omega_p^{-2j}} = \frac{\mathfrak{g}(\phi_{\mathfrak{p}}^{-1})(\sqrt{-D_K})^{r+j} p^{n(-j-r)} \chi_{\mathfrak{p}}^{-1}(p^n)}{(r-1+j)!} \cdot \langle \log_{\mathfrak{p}}(z_{f,\chi}), \omega_f \otimes \omega_A^{r-1+j} \eta_A^{r-1-j} t^{1-2r} \rangle,$$

where Ω_p is the p -adic period of the canonical elliptic curve A in Section 6.1.

In a similar manner to [25, Corollary 6.3], we would like to understand the p -adic valuation of $\langle \log_{\mathfrak{p}}(z_{f,\chi}), \omega_f \otimes \omega_A^{r-1+j} \eta_A^{r-1-j} t^{1-2r} \rangle$. It follows from Theorem 6.1 that we have an inequality

$$v_p \left(\langle \log_{\mathfrak{p}}(z_{f,\chi}), \omega_f \otimes \omega_A^{r-1+j} \eta_A^{r-1-j} t^{1-2r} \rangle \right) \geq n \left(j + r - \frac{1}{2} - v_p(\chi_{\mathfrak{p}}^{-1}(p)) \right)$$

for every anticyclotomic Hecke character ϕ of conductor p^n and infinity type $(j, -j)$ with $-r < j < r$. Here we used the fact that $v_p(\mathfrak{g}(\phi_{\mathfrak{p}}^{-1})) = n/2$ for $n \geq 0$. Under certain hypotheses (see [25, Theorem 5.7], [26, Theorem B]), the μ -invariant $\mu(\mathcal{L}_{\mathfrak{p}}(f))$ vanishes and there is an asymptotic formula:

$$\liminf_{\widehat{\phi}^{-1} \in \mathfrak{X}_{p^\infty}} v_p \left(\langle \log_{\mathfrak{p}}(z_{f,\chi}), \omega_f \otimes \omega_A^{r-1+j} \eta_A^{r-1-j} t^{1-2r} \rangle \right) - n \left(r + j - \frac{1}{2} - v_p(\chi_{\mathfrak{p}}^{-1}(p)) \right) = 0,$$

where p^n is the conductor of ϕ .

We now recall the set-up of Section 5. Let $f_1 \in S_{2r_1}(\Gamma_0(N_1))^{\text{new}}, f_2 \in S_{2r_2}(\Gamma_0(N_2))^{\text{new}}$ be normalized Hecke eigenforms satisfying hypothesis (Heeg). Moreover, suppose that $\chi = \widehat{\phi} \in \mathfrak{X}_{p^\infty}$ is the p -adic avatar of an anticyclotomic Hecke character of infinity type $(j, -j)$ with $-r < j < r$ and conductor $p^n \mathcal{O}_K$ with $n \geq 1$. Let L be a number field containing the Hecke eigenvalues of f_1 and f_2 as well as the values of ϕ , and let $\mathcal{W}$ be the ring of integers of the compositum $L_p \cdot \widehat{\mathbb{Q}_p^{nr}}$. We also denote by $\mathcal{O}_{L_p}$ the ring of integers of L_p . The following Theorem directly follows from Theorem 5.10 and Theorem 6.1.

Theorem 6.2. *Suppose that f_1, f_2 induce isomorphic semi-simplified mod ϖ^m Galois representations: $\bar{\rho}_{f_1}, \bar{\rho}_{f_2} : \text{Gal}(\bar{L}/L) \rightarrow GL_2(\mathcal{O}_{L_p}/\varpi^m \mathcal{O}_{L_p})$, where ϖ is the uniformizer of $\mathcal{W}$ and $\mu(\mathcal{L}_{\mathfrak{p}}(f_1)) = \mu(\mathcal{L}_{\mathfrak{p}}(f_2)) = 0$. Then*

$$\begin{aligned} v_p \left(\left\langle \prod_{\ell|N_1 N_2} \mathcal{P}_{\bar{v}}(f_1)(\chi^{-1}) \log_{\mathfrak{p}}(z_{f_1,\chi}) - \prod_{\ell|N_1 N_2} \mathcal{P}_{\bar{v}}(f_2)(\chi^{-1}) \log_{\mathfrak{p}}(z_{f_2,\chi}), \omega_f \otimes \omega_A^{r-1+j} \eta_A^{r-1-j} t^{1-2r} \right\rangle \right) \\ \geq n \left(j + r - \frac{1}{2} - v_p(\chi_{\mathfrak{p}}^{-1}(p)) \right) + v_p(\varpi^m), \end{aligned}$$

where $\mathcal{P}_{\bar{v}}(f_1), \mathcal{P}_{\bar{v}}(f_2)$ are defined in Definition 5.7.

For a Hecke eigen-newform $f \in S_2(\Gamma_0(N))^{\text{new}}$ of weight 2 satisfying Hypothesis (Heeg) and a finite character χ of conductor p^n , define

$$P(\chi^{-1}) := \sum_{\mathfrak{a} \in \mathcal{C}\ell(\mathcal{O}_{p^n})} \chi(\mathfrak{a})([(A_{\mathfrak{a}}, A_{\mathfrak{a}}[\mathfrak{N}])] - [\infty]) \in J_0(N) \otimes \mathbb{C}_p$$

and let $P_f(\chi^{-1}) := \pi_f(P(\chi^{-1}))$ under the modular parametrization $\pi_f : J_0(N) \rightarrow A_f$. Moreover, let $\omega_{A_f} \in H^0(A_f, \Omega_{A_f}^1)$ be the differential induced by $f(q) \frac{dq}{q} \in$

$H^0(X_0(N), \Omega_{X_0(N)}^1)$ under the Abel-Jacobi map $\iota : X_0(N) \hookrightarrow J_0(N)$ and the projection π_f . In the case of weight 2 forms, we obtain the following extension of [10, Theorem 3.9]:

Theorem 6.3. *Suppose that $f_1 \in S_2(\Gamma_0(N_1))^{new}$, $f_2 \in S_2(\Gamma_0(N_2))^{new}$ induce isomorphic semi-simplified mod ϖ^m Galois representations: $\bar{\rho}_{f_1}, \bar{\rho}_{f_2} : \text{Gal}(\bar{L}/L) \rightarrow GL_2(\mathcal{O}_{L_p}/\varpi^m \mathcal{O}_{L_p})$, where ϖ is a uniformizer of L_p . Then*

$$v_p\left(\prod_{\ell|N_1N_2} \mathcal{P}_{\bar{v}}(f_1)(\chi^{-1})^{\log_{\omega_{A_{f_1}}}(P_{f_1}(\chi^{-1}))} - \prod_{\ell|N_1N_2} \mathcal{P}_{\bar{v}}(f_2)(\chi^{-1})^{\log_{\omega_{A_{f_2}}}(P_{f_2}(\chi^{-1}))}\right) \geq \frac{n}{2} + v_p(\varpi^m).$$

7 Applications to the Iwasawa Main Conjecture for the BDP Selmer group

In this section, we recall the definition of the Bertolini-Darmon-Prasanna (BDP) Selmer group and the corresponding Iwasawa Main Conjecture, which is equivalent to the Heegner Point Main Conjecture formulated by Perrin-Riou [27]. We will see that the Iwasawa Main Conjecture propagates in a family of modular forms with isomorphic semi-simplified residual representations.

We give the following definitions based on [28, Definitions 2.1, 2.2], [29, p.98]. Assume that $f \in S_{2r}(\Gamma_0(N))^{new}$ is ordinary at p , i.e. $a_p(f) \in \mathbb{Z}_p^\times$. Recall the p -adic representation $V = V_f(r)$ of $\text{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p)$ attached to f . There exists a $\text{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p)$ -stable filtration

$$0 \rightarrow \mathcal{F}^+V \rightarrow V \rightarrow \mathcal{F}^-V \rightarrow 0$$

where $\mathcal{F}^+V$ and $\mathcal{F}^-V$ are both 1-dimensional representations. Let T be a G_F -stable lattice in V and let $A = V/T$. We also define $\mathcal{F}^+T = T \cap \mathcal{F}^+V$, $\mathcal{F}^-T = T/\mathcal{F}^+T$, and $\mathcal{F}^+A = \mathcal{F}^+V/\mathcal{F}^+T$, $\mathcal{F}^-A = A/\mathcal{F}^+A$.

To define the BDP Selmer group, we recall the following local conditions above p , where M is A, V or T . Let F/K be a finite extension, and let v be a prime of F .

Definition 7.1. The Greenberg local condition is defined as

$$H_{\text{Gr}}^1(F_v, M) := \begin{cases} \ker(H^1(F_v, M) \rightarrow H^1(F_v^{nr}, \mathcal{F}^-M)) & \text{if } v \mid p, \\ \ker(H^1(F_v, M) \rightarrow H^1(F_v^{nr}, M)) & \text{if otherwise.} \end{cases}$$

Definition 7.2. For $v \mid p$ and $\mathcal{L}_v \in \{\emptyset, \text{Gr}, 0\}$, set

$$H_{\mathcal{L}_v}^1(F_v, M) := \begin{cases} H^1(F_v, M) & \text{if } \mathcal{L}_v = \emptyset, \\ H_{\text{Gr}}^1(F_v, M) & \text{if } \mathcal{L}_v = \text{Gr}, \\ \{0\} & \text{if } \mathcal{L}_v = 0. \end{cases}$$

Let Σ be a finite set of primes of K dividing the primes where V is ramified as well as the primes dividing $p\infty$. We will denote by F_Σ the maximal extension of F unramified outside of the set of primes dividing the primes in Σ .

Definition 7.3. For a set of local conditions $\mathcal{L} = \{\mathcal{L}_v\}_{v|p}$, we define

$$\mathrm{Sel}_{\mathcal{L}}(F, M) = \ker \left(H^1(F_{\Sigma}/F, M) \rightarrow \prod_{v \nmid p} \frac{H^1(F_v, M)}{H_{\mathrm{Gr}}^1(F_v, M)} \times \prod_{v|p} \frac{H^1(F_v, M)}{H_{\mathcal{L}_v}^1(F_v, M)} \right).$$

We abbreviate the Iwasawa algebras as $\Lambda := \mathbb{Z}_p[[\Gamma_K^-]]$, $\Lambda^{\mathrm{ur}} := \mathcal{W}[[\Gamma_K^-]]$ and define $\mathbf{T} := T \otimes \Lambda$, and $\mathbf{A} := A \otimes \Lambda^*$, where Λ^* is the Pontryagin dual of Λ .

Observe that there are isomorphisms

$$\mathrm{Sel}_{\mathcal{L}}(K_{\infty}, \mathbf{A}) := \varinjlim_{K \subset F \subset K_{\infty}} \mathrm{Sel}_{\mathcal{L}}(F, A), \quad \mathrm{Sel}_{\mathcal{L}}(K_{\infty}, \mathbf{T}) := \varprojlim_{K \subset F \subset K_{\infty}} \mathrm{Sel}_{\mathcal{L}}(F, T)$$

with compatible local conditions $\mathcal{L}$. We will also denote by $X_{\mathcal{L}}(K, \mathbf{A})$ the Pontryagin dual of $\mathrm{Sel}_{\mathcal{L}}(K, \mathbf{A})$.

The BDP Selmer group is defined as $\mathrm{Sel}_{\emptyset, 0}(K, \mathbf{A})$, and we will denote its dual by $X_{\mathfrak{p}}(K, \mathbf{A}) := X_{\emptyset, 0}(K, \mathbf{A})$. One can formulate the following Iwasawa main conjecture [30, Conjecture 2.4.7]:

Conjecture 7.4 (Iwasawa main conjecture). *The BDP Selmer group $X_{\mathfrak{p}}(K, \mathbf{A})$ is $\mathbb{Z}_p[[\Gamma_K^-]]$ -cotorsion, and*

$$\mathrm{char}(X_{\mathfrak{p}}(K, \mathbf{A})) \otimes_{\mathbb{Z}_p[[\Gamma_K^-]]} \mathcal{W}[[\Gamma_K^-]] = (\mathcal{L}_{\mathfrak{p}}(f)^2)$$

as ideals in $\mathcal{W}[[\Gamma_K^-]]$, where $\mathrm{char}(X_{\mathfrak{p}}(K, \mathbf{A}))$ is the characteristic ideal of $X_{\mathfrak{p}}(K, \mathbf{A})$.

We also remark that Conjecture 7.4 is equivalent to Perrin-Riou's Heegner Point Main Conjecture for forms corresponding to elliptic curves. For more details on this equivalence, we refer readers to [18, 27].

Denote by $\mu_{\mathrm{anal}}(f)$ and $\lambda_{\mathrm{anal}}(f)$ the μ and λ -invariants of $\mathcal{L}_{\mathfrak{p}}(f)$, respectively. Moreover, let $\mu_{\mathrm{alg}}(f) := \mu(X_{\mathfrak{p}}(K, \mathbf{A}))$ and $\lambda_{\mathrm{alg}}(f) := \lambda(X_{\mathfrak{p}}(K, \mathbf{A}))$ be the algebraic μ and λ -invariants of $X_{\mathfrak{p}}(K, \mathbf{A})$. Combining our results with the work of Lei-Mueller-Xia [9], we obtain the following:

Theorem 7.5. *Let $f_1 \in S_{2r_1}(\Gamma_0(N_1))^{\mathrm{new}}$ be a newform that satisfies Conjecture 7.4 and assume $\mu_{\mathrm{anal}}(f_1) = \mu_{\mathrm{alg}}(f_1) = 0$. Suppose that $f_2 \in S_{2r_2}(\Gamma_0(N_2))^{\mathrm{new}}$ is a newform that satisfies the divisibility*

$$\mathcal{L}_{\mathfrak{p}}(f_2)^2 \in \mathrm{char}_{\Lambda}(X_{\mathfrak{p}}(K, \mathbf{A}_2)),$$

where $X_{\mathfrak{p}}(K, \mathbf{A}_i)$ is the dual BDP Selmer group for f_i , $i \in \{1, 2\}$. Further suppose that $\bar{\rho}_{f_1} \simeq \bar{\rho}_{f_2} \pmod{\varpi}$ and $H^0(K_w, A_i) = 0$ for every $w \mid p$ and $i \in \{1, 2\}$. Then $\mu_{\mathrm{anal}}(f_2) = \mu_{\mathrm{alg}}(f_2) = 0$ and Conjecture 7.4 also holds for f_2 .

Proof. Under these hypotheses, Theorem 5.10 and [9, Corollary 3.8] imply that $\mu_{\mathrm{alg}}(f_2) = \mu_{\mathrm{anal}}(f_2) = 0$.

Moreover, we also have

$$2\lambda(\mathcal{L}_{\mathfrak{p}}(f_1)) + 2 \sum_{\ell | N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_1)) = 2\lambda(\mathcal{L}_{\mathfrak{p}}(f_2)) + 2 \sum_{\ell | N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_2)) \quad (7.1)$$

for any splitting $\ell = v\bar{v}$ in K of the primes $\ell \mid N_1 N_2$. For $i \in \{1, 2\}$, each $\mathcal{P}_{\bar{v}}(f_i)$ is defined in Definition 5.7.

By [9, Corollary 3.8], one also has

$$\lambda(X_{\mathfrak{p}}(K, \mathbf{A}_1)) + 2 \sum_{\ell | N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_1)) = \lambda(X_{\mathfrak{p}}(K, \mathbf{A}_2)) + 2 \sum_{\ell | N_1 N_2} \lambda(\mathcal{P}_{\bar{v}}(f_2)). \quad (7.2)$$

Conjecture 7.4 for f_1 gives $\lambda_{\text{anal}}(f_1) = \lambda_{\text{alg}}(f_1)$. The equalities (7.1) and (7.2) together imply that $\lambda_{\text{anal}}(f_2) = \lambda_{\text{alg}}(f_2)$. Combined with the divisibility for f_2 , we conclude that Conjecture 7.4 also holds for f_2 . $\square$

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